

MANGELEK GABRIEL NICHOLAS TAMBUN

Data Engineer | Machine Learning Engineer | Full-Stack Developer

Semarang, Central Java, Indonesia | Open to Jakarta, Surabaya, Bali | +62 813-7004-4147 | nicholas.tbn26@gmail.com

[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Informatics graduate from Universitas Diponegoro with hands-on experience across data engineering, machine learning, and full-stack web development. Built financial data pipelines for 3 state-owned entities, developed an LLM-enhanced stock forecasting system, and delivered dashboard applications. Strong foundation in data workflow design, predictive modeling, system integration, and production-oriented dashboard development.

EDUCATION

Universitas Diponegoro — Semarang, Indonesia

Aug 2022 – Jun 2026

Bachelor of Informatics, GPA: 3.52/4.00

- Thesis: Developed a multi-horizon stock price forecasting system for Indonesian blue-chip stocks using Temporal Fusion Transformer and LLM-based news sentiment features.

PROFESSIONAL EXPERIENCE

Freelance Web Developer — ICU Monitoring SIMRS

Oct 2025 – Dec 2025

Prima Insan Mulia Hospital — Remote · Freelance

- Built the frontend of a 24-hour ICU patient-monitoring system using Next.js, React, TypeScript, and Tailwind CSS, covering 5 clinical modules: hemodynamic monitoring, clinical instructions, invasive equipment, nurse planning, and patient progress tracking.
- Integrated 26 service-layer API calls across 7 modules to deliver real-time per-bed patient data and reduce dependency on manual paper-based tracking.

Data System Engineer — Internship

Jan 2025 – Mar 2025

PT Biro Klasifikasi Indonesia (BKI) — Tanjung Priok, Jakarta · On-site

- Built 12-stage PySpark ETL pipelines to integrate monthly financial data from 3 state-owned entities: BKI, SUCOFINDO, and Surveyor Indonesia.
- Transformed chart-of-account and entity-level financial data using joins, grouping, and pivot operations to support Balance Sheet and Profit & Loss reporting.
- Developed COA-to-Balance Sheet and COA-to-Profit & Loss mapping modules using PySpark, PostgreSQL, SQL, and JDBC, reducing dependency on manual report compilation.

Junior Software Engineer — Internship

Jan 2025 – Mar 2025

Aluesa Global Digitek — Remote · Odoo ERP Development

- Configured 5 Odoo ERP modules — accounting, invoicing, asset management, expenses, and reporting — to support daily internal operations.
- Migrated financial records across 3 categories (assets, expenses, accounting) into Odoo, verifying data integrity post-migration.
- Built dashboard reporting filters and data-mapping workflows to streamline report generation.

Laboratory Assistant

Aug 2023 – Jul 2024

Diponegoro University — Semarang, Indonesia · On-site, Contract

- Taught programming fundamentals and algorithms to 30 undergraduate students across 2 academic terms, leading 11 lab sessions per term.
- Supported assignment completion for 30 students through structured mentoring and code review.

PROJECTS

Temporal Fusion Transformer Stock Forecasting System

Oct 2025 – Jan 2026

Undergraduate Thesis — Diponegoro University

- Built a TFT model for multi-horizon forecasting of 4 Indonesian blue-chip stocks (ASII, BBRI, BMRI, TLKM) using 9 years of historical price data.
- Developed an LLM-enhanced TFT variant with news-sentiment features, improving MAPE from 2.26% (baseline) to 2.13%.

- Achieved 58.3% Directional Accuracy and R^2 of 0.988 across evaluated tickers.

Financial Consolidation Dashboard

Jan 2025 – Apr 2025

Independent extension of the PT Biro Klasifikasi Indonesia (BKI) internship — built a reporting front end

- Designed and built a full-stack reporting front end (React.js, Material UI, Express.js) consuming the internship's consolidated PostgreSQL data, with authenticated views for 3 entity stakeholders.
- Designed line/bar/pie chart visualizations for key financial KPIs, supporting monthly review meetings across all 3 entities.

ABSENSI — Face Recognition Attendance Dashboard

Nov 2025

- Built a real-time face/identity recognition dashboard (YOLOv8, Streamlit) covering 16 identity classes across image, batch, video, and webcam inputs.
- Added 3 configurable inference controls (confidence, IoU, image size) with CSV-exportable detection logs.

Skills & Interests

Technical: Python, SQL, TypeScript, PySpark, PostgreSQL, React.js, Next.js, Node.js

Language: Indonesian — Native; English — Professional Working Proficiency

Interests: Data Engineering, Machine Learning, Full-Stack Development, Financial Technology

CERTIFICATIONS & TRAINING

- Data Science 101 — Cognitive Class (IBM) | Mar 2025
- Data Science Methodology — Cognitive Class (IBM) | Mar 2025
- ACA Cloud Computing Certification — Alibaba Cloud | Nov 2024 (Valid through Nov 2026)
- Belajar Dasar Visualisasi Data — Dicoding Indonesia | Apr 2024 (Valid through Apr 2027)
- Database Design — Oracle | Dec 2023
- Data Analysis: Full-Stack Intensive Bootcamp — MySkill | Oct 2023